



Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
Second Semester 2025-2026

Digital Learning Handout

Part A: Content Design

Course Title	AI and ML Techniques in Cyber Security
Course No(s)	AIMLC ZG567
Credit Units	5
Credit Model	3-1-1
Course Author	Prof. Seetha Parameswaran
Instructor	Prof Sashank Dara
Version No:	1
Date:	27/02/2025

Course Description:

Introduction to Cyber-Security; Supervised Learning for Misuse/Signature Detection; Machine Learning for Anomaly Detection; Malware detection and classification; Network Intrusion detection and classification; Detection and categorization of domain names; Profiling Network Traffic; Adversarial Machine Learning for Malware detection

Course Objectives

No	Course Objective
CO1	Apply machine learning principles to cybersecurity challenges through a structured approach that integrates traditional security concepts with data science methodologies across various threat detection domains.
CO2	Analyze security data using appropriate machine learning algorithms to identify patterns, anomalies, and potential threats in different contexts including network traffic, malware, and domain names.
CO3	Evaluate the effectiveness of machine learning models for security applications by assessing their performance, limitations, and vulnerabilities against both conventional and adversarial threats.
CO4	Design comprehensive security solutions that leverage machine learning capabilities to enhance threat detection, reduce false positives, and improve overall security posture within modern cybersecurity frameworks.

Text Book(s):

T1	Chio, C., & Freeman, D. (2018). Machine learning and security: Protecting systems with data and algorithms. O'Reilly Media, Inc..
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Reference Book(s) & other resources:

R1	Stallings, W., & Brown, L. (2015). Computer security: principles and practice. Pearson.
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R2	Mitchell, T. M., & Mitchell, T. M. (1997). Machine learning (Vol. 1, No. 9). New York: McGraw-hill.
R3	Soma, H., & Sinan, O. (2018). Hands-On Machine Learning for Cybersecurity. Birmingham–Mumbai: Packt Publishing.
R4	Dua, S., & Du, X. (2016). Data mining and machine learning in cybersecurity. CRC press.

Learning Outcomes: Students will be able to

LO1	Apply machine learning techniques to analyze cyber threats by implementing supervised and unsupervised models for signature detection and anomaly detection.
LO2	Analyze network traffic patterns using appropriate feature extraction methods and classification algorithms to identify potential intrusions, evaluating model performance using standard security metrics.
LO3	Evaluate the effectiveness of different machine learning approaches for malware detection by comparing their performance on real-world datasets, assessing their strengths and limitations against evasion techniques, and documenting findings in a comprehensive technical report.
LO4	Design robust defense strategies against adversarial machine learning attacks by implementing model hardening techniques, developing detection mechanisms for poisoning and evasion attempts, and creating a security framework that addresses the ML pipeline vulnerabilities.

Modular Content Structure

1. Introduction to Cyber Security
 - Fundamentals of Cybersecurity
 - AI/ML in Cybersecurity: Landscape
2. Core Cybersecurity Concepts
 - Security Architecture and Defense Strategies
 - Threat Intelligence and Attribution
 - Tactics, techniques, and procedures
 - Security Operations Center (SOC)
3. Data Science and ML Foundations for Cybersecurity
 - Data Collection and Preprocessing
 - ML Algorithms for Security Applications
 - ML Development Pipeline for Security
4. Supervised Learning for Misuse/Signature Detection
 - Signature-based Detection Principles
 - Supervised Learning Techniques (Decision Trees, SVM, Random Forest)
 - Case Study: Building a Classifier
5. Machine Learning for Anomaly Detection
 - Anomaly Detection Fundamentals
 - Unsupervised Learning for Anomaly Detection (K-means, DBSCAN)
 - Real-time Anomaly Detection Systems
6. Malware Detection and Classification





- Malware Characteristics and Types
- ML-based Malware Detection
- Deep learning for malware classification
- Evasion Techniques and Countermeasures
- 7. Network Intrusion Detection and Classification
 - Network Traffic Analysis Basics
 - ML Models for Network Intrusion Detection
 - Evaluation Using Standard Datasets NSL-KDD
- 8. Detection and Categorization of Domain Names
 - Domain Generation Algorithms (DGAs)
 - ML Techniques for Domain Classification
- 9. Profiling Network Traffic
 - Traffic Profiling Methods
 - ML for Traffic Classification
- 10. Adversarial Machine Learning for Malware Detection
 - Adversarial Machine Learning Fundamentals
 - Attack types (evasion, poisoning, model stealing)
 - Defense Strategies

Part B: Learning Plan

Contact Session	List of Topic Title	Sub-Topics	Reference
1	Introduction to Cyber Security	● Fundamentals of Cybersecurity ○ Core concepts and terminology ○ CIA triad (Confidentiality, Integrity, Availability) ○ Types of cyber threats and attack vectors ● AI/ML in Cybersecurity: Landscape ○ Evolution of AI in security applications ○ Challenges and opportunities ○ Key application areas in security	T1 - Ch 1
2	Core Cybersecurity Concepts	● Security Architecture and Defense Strategies ○ Defense-in-depth approach ○ Security policies and frameworks ○ Attack surface management ● Threat Intelligence and Attribution ○ Threat actors and motivations ○ Tactics, techniques, and procedures (TTPs) ○ Threat modeling approaches ● Security Operations Center (SOC) ○ SOC structure and functions ○ Detection and response workflows ○ Integration of ML in modern SOC's	R1





3	Data Science and ML Foundations for Cybersecurity	• Data Collection and Preprocessing ○ Security data sources and formats ○ Data cleaning techniques for security datasets ○ Feature extraction challenges in security contexts • ML Algorithms for Security Applications ○ Algorithm selection criteria for security tasks ○ Supervised vs. unsupervised approaches ○ Model evaluation in security contexts • ML Development Pipeline for Security ○ Data labeling challenges ○ Model deployment considerations ○ Maintaining model effectiveness over time	T1 - Ch 2
4	Supervised Learning for Misuse/Signature Detection	• Signature-based Detection Principles ○ Traditional signature development ○ Pattern matching algorithms ○ Limitations of signature-based approaches • Supervised Learning Techniques ○ Classification algorithms (Decision Trees, SVM, Random Forest) ○ Feature engineering for signature detection ○ Model evaluation metrics • Case Study: Building a Classifier ○ Dataset exploration and preparation ○ Model training and optimization ○ Performance analysis and interpretation	R4 - Ch 3
5	Machine Learning for Anomaly Detection	• Anomaly Detection Fundamentals ○ Statistical methods ○ Distance-based methods ○ Density-based methods • Unsupervised Learning for Anomaly Detection ○ Clustering approaches (K-means, DBSCAN) ○ Dimensionality reduction (PCA, autoencoders) ○ One-class classification methods • Real-time Anomaly Detection Systems ○ Implementation challenges ○ Threshold optimization ○ Reducing false positives	T1 - Ch 3
6	Malware Detection and Classification	• Malware Characteristics and Types ○ Evolution of malware ○ Static vs. dynamic analysis ○ Feature extraction from malware samples • ML-based Malware Detection	T1 - Ch 4





		○ N-gram analysis ○ API call sequence analysis	
7	Malware Detection and Classification	● Deep learning for malware classification ● Evasion Techniques and Countermeasures ○ Obfuscation methods ○ Polymorphic and metamorphic malware ○ Model adaptation strategies	T1 - Ch 4
8	Review	Review of contact session 1 to 7	
9	Network Intrusion Detection and Classification	● Network Traffic Analysis Basics ○ Protocol analysis ○ Flow-based features ○ Packet-level inspection ● ML Models for Network Intrusion Detection ○ Feature selection for network data ○ Ensemble methods for improved detection	T1 - Ch 5
10	Network Intrusion Detection and Classification	● Real-time classification approaches ● Evaluation Using Standard Datasets ○ NSL-KDD, CICIDS, and other benchmarks ○ Performance metrics specific to NIDS ○ Operational considerations	T1 - Ch 5
11	Detection and Categorization of Domain Names	● Domain Generation Algorithms (DGAs) ○ Characteristics of malicious domains ○ Linguistic and statistical features ○ Time-based patterns ● ML Techniques for Domain Classification ○ N-gram models ○ Deep learning approaches ○ Real-time detection systems	Class notes
12	Profiling Network Traffic	● Traffic Profiling Methods ○ Statistical fingerprinting ○ Protocol behavior analysis ○ User and entity behavior analytics	R4 - Ch 7
13	Profiling Network Traffic	● ML for Traffic Classification ○ Application identification ○ Encrypted traffic analysis ○ Temporal pattern recognition	R4 - Ch 7
14	Adversarial Machine Learning	● Attack types (evasion, poisoning, model stealing) ○ Vulnerability of ML models in security context	T1 - Ch 8





	Fundamentals	○ Adversarial examples in malware ● Defense Strategies ○ Adversarial training ○ Model robustness techniques ○ Detection of adversarial inputs	
15	Adversarial Machine Learning Fundamentals	● Attack Technique: Model Poisoning ○ Example: Binary Classifier Poisoning Attack ○ Attacker Knowledge ○ Defense Against Poisoning Attacks ● Attack Technique: Evasion Attack ○ Example: Binary Classifier Evasion Attack ○ Defense Against Evasion Attacks	T1- Ch 8
16	Review	Review of contact session 9 to 15	

Experiential Learning Components:

1. Lab work: 6
2. Project work: 0
3. Case Study: 0
4. Simulation: 0
5. Work Integrated Learning Assignment- 2 Assignments
6. Design work/ Field work: 0

Objective of Experiential Learning Component:

Hands on sessions on implementation of

- a) Malware detection and classification
- b) Network Intrusion Detection and Classification

Scope of Experiential Learning Component:

Programming language: Python

Tools and libraries: Jupyter, Numpy, Scipy, Pandas, ScikitLearn, Matplotlib, Seaborn, Tensorflow, Keras

Lab Infrastructure:

Online / Open source / Virtual lab / Google Colab

List of Experiments:

Lab No	Lab Objective	Session Reference
1	Misuse/Signature Detection	3
2	Anomaly Detection	4
3	Malware detection	6





4	Malware classification	7
5	Network Intrusion Detection and Classification	9,10
6	Profiling Network Traffic	12. 13

Evaluation Scheme:

Legend: EC = Evaluation Component; AN = After Noon Session; FN = Fore Noon Session

Evaluation Component	Name (Quiz, Lab, Project, Mid-term exam, End semester exam, etc.)	Type (Open book, Closed book, Online, etc.)	Weight	Duration	Day, Date, Session, Time
EC – 1*	Quiz	Online	10%	1 week	TBA
	Assignment/Lab Assignment / Lab Exams	Online	20 %	10 days	TBA
EC - 2	Mid-Semester Test	Closed Book	30%	2 hours	20-06-2026 (AN)
EC - 3	Comprehensive Exam	Open Book	40%	2 ½ Hours	05-09-2026 (AN)

EC1* (20% - 30%): Quiz (optional): 5-10 %, Lab Assignment/Assignment: 20% - 30%

Syllabus for Mid-Semester Test (Closed Book): Topics in Contact session: 1 to 8

Syllabus for Comprehensive Exam (Open Book): All topics

Important Links and Information:

eLearn Portal: <https://elearn.bits-pilani.ac.in>

Students must visit the eLearn portal regularly and stay updated with the latest announcements and deadlines.

Contact Sessions: Students should attend the online lectures as per the schedule provided on the eLearn portal.

Evaluation Guidelines:

1. EC-1 consists of either two Assignments or three Quizzes. Students will attempt them through the course pages on the eLearn portal. Announcements will be made on the portal in a timely manner.
2. For Closed Book tests: No books or reference material of any kind will be permitted.
3. For Open Book exams: “open book” means text/ reference books (publisher copy only) and does not include any other learning material. No other learning material will be permitted during the open book examinations. For Detailed Guidelines refer to the attached document.

[EC3 Guidelines](#)

4. If a student is unable to appear for the Regular Test/Exam due to genuine exigencies, the student should follow the procedure to apply for the Make-Up Test/Exam, which will be made available on the eLearn portal. The Make-Up Test/Exam will be conducted only at selected exam centres on the dates to be announced later.

It shall be the responsibility of the individual student to be regular in maintaining the self-study schedule as given in the course handout, attend the online lectures, and take all the prescribed evaluation components such as Assignments/Quizzes, Mid-Semester Tests and Comprehensive Exams according to the evaluation scheme provided in the handout.

